Glass Box Trading

AI proposes; deterministic gates dispose.

- Result at 2026-09-03T20:00:14.787Z: $583.59 (0.58%) on a $100k paper account
- Every decision — trade or no trade — is public and broker-reconciled
- Demo: https://glass-box-trading.vercel.app · Repo: https://github.com/fradzano/glass-box-trading

The auditability problem

- "Autonomous trading agent" claims are easy to make, hard to verify
- Target user: a trader or allocator who wants unattended execution but distrusts opaque bot claims
- This entry publishes the whole decision trail — proposals, vetoes, and fills alike — not just the wins

The decision-to-fill golden path

1. Open the dashboard, no login — see account state and control model
2. Pick one decision cycle — market context, candidate, rationale
3. See the full gate vector, including at least one veto
4. Follow one approved intent to its order, fill, and P&L contribution
5. Open the public source at the pure decision core and its tests

"AI proposes; deterministic gates dispose"

- LLM analyst proposes candidates from read-only Alpaca market data (MCP)
- Analyst output is schema-validated and whitelist-constrained — never a free-form order
- A pure, tested decision core owns every risk gate; time/config/account state are parameters, not ambient calls
- Executor places only core-approved actions via Alpaca CLI/API — the LLM has no code path to an order

Options strategy and declared variance allocation

- Two-sleeve barbell, declared up front, not fit after the fact:
  - Reserve ~80% — cash, idle
  - Income ~12% (max-loss basis) — defined-risk credit spreads / iron condors
  - Convex ~8% (premium basis) — debit spreads / long options
- Defined-risk only, everywhere — no naked short options in the strategy space

Risk gates and bounded unattended worst case

- Entry gates: defined-risk only, sleeve budgets, per-position max loss, concentration cap, liquidity floor, session gate, idempotency, schema gate
- Lifecycle gates: expiry eviction, assignment reconciliation, deadline flatten by Sep 3 close
- State/failure gates: single-instance epoch + halt flag, drawdown kill-switch, dead-man watchdog
- Worst case is bounded by design, not by monitoring diligence

Broker-reconciled P&L and sleeve attribution

- Account PA376WIK2ATL, journal revision sha256:7b82959a344a7c7e
- Start equity $100,000.00 · Equity at cutoff $100,583.59 · P&L $583.59 (0.58%) at 2026-09-03T20:00:14.787Z
- Realized $586.00 · Unrealized $0.00 · Unattributed -$2.41 — a fee-shaped residual the journal cannot explain, shown and never assigned to a sleeve
- Realized only, by sleeve: income $230.00 · convex $356.00
- Max drawdown $33.15 (0.03% of peak $100,596.19); book flat, zero positions, zero orders

Public journal, vetoes, and originality

- Every cycle's proposal or no-trade result is journaled with its gate vector and rationale — rejections logged with the same fidelity as fills
- The declared variance split is the originality claim, not a marketed edge
- 29 vetoed candidates and 24 no-trade cycles are visible on the dashboard as of 2026-09-03T20:00:14.787Z

Failure drills, two defects found, and limitations

- Failure-path tests cover reconciliation, kill-switch, watchdog, expiry
- **Defect 1:** on flatten day the runner could not price the structures expiring next session (quote window starts at `EXPIRY_MIN_SESSIONS`); the refusal reached only a discarded report, so no close was submitted
- **Response:** owner stood the writer down, the certified dead-man watchdog took over — `HALT WATCHDOG_TAKEOVER`, all three structures closed and filled within a second, book flat at the deadline
- **Defect 2:** that watchdog's wrapper had killed the CLI on its first stderr line since arming — inert for a day, fixed at 17:41 CEST, an hour before the takeover
- **API limit:** no conditional submit at Alpaca; the pre-submit re-fetch is the declared linearization point, a manual mutation halts the next cycle
- Paper only; one week of paper P&L cannot prove edge — no alpha claim

Demo, repository, and post-hackathon path

- Demo: https://glass-box-trading.vercel.app
- Repository: https://github.com/fradzano/glass-box-trading
- Journal revision at cutoff: sha256:7b82959a344a7c7e
- Post-hackathon: continue paper operation, extend the gate catalog and sleeve set before any live-account discussion